

Vikas Verma

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Skills

Programming Languages: Python, JavaScript

AI / ML: Generative AI, LLMs, RAG, AI Agents, NLP, Deep Learning (CNN, LSTM)

Frameworks & Libraries: PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face Transformers, FastAPI, Flask, OpenCV

Cloud & Infrastructure: AWS(S3, EC2, ECS, DynamoDB, Lambda, Bedrock, API Gateway, IAM, CloudWatch), Docker, Terraform

MLOps & Deployment: MLflow, DVC, Model Deployment, Model Monitoring, Experiment Tracking

Databases & Vector Stores: MongoDB, MySQL, QdrantDB

Computer Vision: Object Detection, OCR, Image Segmentation, Human Action Recognition

GenAI Tooling: LangChain, LangGraph, Multi-Agent Systems, MCP, CrewAI, AutoGen, Agentic AI Workflows, Tool Calling, Embedding Models.

Work Experience

Wittybrains Software Technologies Private Limited

Oct 2024 – Present

AI / ML Engineer

- Designed and deployed an enterprise-grade RAG chatbot processing 10K+ documents, improving query accuracy by 35% and reducing manual search time by 60%.
- Built scalable AWS-based pipelines for document ingestion, embedding generation, and vector retrieval, enabling efficient large-scale semantic search.
- Implemented intelligent query classification and prompt routing to dynamically switch between general LLM responses and document-grounded RAG workflows.
- Developed and deployed real-time computer vision pipelines using YOLOv10 for object detection.
- Built end-to-end machine learning pipelines for human action recognition (CNN + LSTM) and image segmentation tasks.
- Trained and deployed deep learning models using TensorFlow and PyTorch, built autonomous AI agents and orchestrated multi-agent workflows using LangChain and LangGraph, and leveraged MLflow, DVC, and Docker for experiment tracking, model versioning, and containerized deployment.

Education

IIMT College of Engineering, Greater Noida

Master of Computer Applications (MCA)

2022 – 2024

Percentage: 74%

Goel Institute of Higher Studies, Lucknow

Bachelor of Computer Applications (BCA)

2019 – 2022

Percentage: 77%

Project Work

Enterprise RAG Chatbot

- Built an enterprise-grade RAG chatbot for document-based question answering.
- Developed scalable ingestion pipelines, embedding generation, and vector search workflows.
- Implemented dynamic prompt routing between LLM and document-grounded responses.
- Integrated FastAPI services with logging, monitoring, and cloud deployment.
- Used LangChain and QdrantDB for orchestration and vector similarity search.

Human Action Recognition System

- Developed a CNN + LSTM based activity recognition system on drone video streams.
- Built preprocessing, feature extraction, and temporal sequence modeling pipelines.
- Developed and trained deep learning models using TensorFlow.
- Improved model performance using hyperparameter tuning and data augmentation.
- Optimized inference pipelines for near real-time activity prediction.

Image Segmentation & Quality Assessment

- Developed a CNN-based saffron quality assessment and image segmentation system.
- Built preprocessing, augmentation, and automated model training workflows.
- Trained deep learning models for segmentation and classification tasks.
- Applied MLOps practices including MLflow, experiment tracking, and versioning.
- Implemented OpenCV-based image preprocessing and feature enhancement techniques.